



College ERP System

Complete Enterprise Technical Architecture, File Directory Layout & Role Workflows

ARCHITECTURE: 3-TIER ENTERPRISE

FRONTEND: REACT 18 + VITE

BACKEND: JAVA SERVLETS + REST

DATABASE: ORACLE 21C

1. High-Level System Architecture

Client / Presentation

FRONTEND

- React 18 SPA with Vite
- Bootstrap 5 + Vanilla CSS
- Responsive Role Dashboards
- Client State & Notifications
- Axios HTTP Client & Router

Application Layer

BACKEND

- Java 17/21 Servlets (Jakarta API)
- RESTful JSON Controllers
- Apache Tomcat 10+ Container
- DAO Data Access Pattern
- BCrypt Password Hashing

Data Storage Layer

DATABASE

- Oracle Database 21c XE
- PL/SQL Triggers & Sequences
- Foreign Key Integrity
- OJDBC8 Connection Pooling
- Role-Based Access Tables

2. Complete Project Directory Layout

```
ERP Files/
├── README.md                # Project documentation and quickstart guide
├── .gitignore               # Git rules excluding build artifacts and secrets
├── database/                # Oracle SQL scripts
│   ├── schema.sql          # DDL: Tables, constraints, sequences, triggers
│   └── seed_data.sql        # DML: Demo users, departments, subjects, books
├── backend/                 # Java Maven Web Application
│   ├── pom.xml              # Maven dependencies: OJDBC8, Jackson, BCrypt
│   └── src/main/java/com/collegeerp/
│       ├── config/          # DBConnection.java, CORSFilter.java
│       ├── controller/      # AuthServlet, AttendanceServlet, FeeServlet, LibraryServlet
│       ├── dao/              # UserDAO, AttendanceDAO, FeeDAO, LibraryDAO
│       ├── model/            # User, Attendance, Fee, Assignment, Book
│       └── util/              # JSONResponse, PasswordUtil
├── frontend/                # React 18 + Vite Web Application
│   ├── package.json          # Node dependencies: react-router-dom, bootstrap, react-icons
│   ├── vite.config.js        # Vite build and dev server configuration
│   ├── public/               # Static assets and SPA _redirects rule
│   └── src/
│       ├── components/common/ # Navbar, Collapsible Sidebar, StatCard, Footer
│       ├── components/modals/ # ApplyLeaveModal, BookActionModal
│       ├── context/           # AuthContext (user session, role state)
│       ├── routes/            # AppRoutes (Protected role-based route matrix)
│       ├── services/          # authService (demo mode & API), api.js
│       ├── shared-modules/     # LeaveRequestsView, AttendanceView, LibraryView, etc.
│       └── views/              # Role Dashboards (Student, Faculty, HOD, Principal, etc.)
```

3. Role-Based Module Permissions & Routing

ROLE	DASHBOARD PATH	PERMITTED NAVIGATION MODULES	RESTRICTED / HIDDEN MODULES
Student	/student/dashboard	Attendance, Timetable, Assignments, Fee Status, Library, Leave Request, Notices	User Management, Review Approvals
Faculty	/faculty/dashboard	Attendance Marking, Timetable, Assignments Hub, Student Leave Approvals, Library, Notices	Fee Status, User Management
HOD	/hod/dashboard	Department Metrics, Faculty Management, Faculty Leave Approvals, Subjects, Library, Notices	System User Deletion
Principal	/principal/dashboard	College Overview, HOD/Staff Approvals, Institutional Notices, Library Catalog	Timetable Scheduling, Assignment Creation
Accounts	/accounts/dashboard	Fee Collection, Student Invoices, Staff Leave Applications, Notices	Timetable, Assignments, Subjects
Library Staff	/library/dashboard	Book Inventory, Circulation Desk (Issue/Return), Staff Leave Applications, Notices	Attendance, Timetable, Assignments
System Admin	/admin/dashboard	User Management (Create/Delete Users), System Audit, System Announcements	Attendance Marking, Timetable, Assignments

4. Hierarchical Leave Approval & Notification Flow

- 1 Student Leave Submission → Class Advisor / Faculty Review

Students apply with leave type, dates, and reason. The request automatically routes to **Faculty** who can Approve or Reject from their dedicated Approvals tab.
- 2 Faculty Leave Submission → Head of Department (HOD) Review

Faculty members submit leaves that automatically route to their department's **HOD** (e.g. Computer Science HOD or Chemical HOD) for departmental review.
- 3 HOD, Accounts & Library Staff → Principal & Director Review

Department heads and institutional staff submit leaves that escalate directly to the **Principal** for executive clearance.
- 4 Instant Notification & Status Update Dispatch

Upon clicking Approve or Reject, an immediate notification is pushed to the applicant's notification bell panel and synchronized across dashboards.

5. Core Database Relational Entities

USERS Table • USER_ID (PK, NUMBER) • USERNAME, PASSWORD_HASH • FULL_NAME, EMAIL, PHONE_NUMBER • ROLE (STUDENT, FACULTY, HOD, etc.) • DEPARTMENT_ID (FK → DEPARTMENTS)	ATTENDANCE Table • ATTENDANCE_ID (PK, NUMBER) • STUDENT_ID (FK → USERS) • SUBJECT_ID (FK → SUBJECTS) • STATUS ('PRESENT', 'ABSENT') • ATTENDANCE_DATE (DATE)
FEES Table • FEE_ID (PK, NUMBER) • STUDENT_ID (FK → USERS) • TOTAL_FEE, PAID_AMOUNT • DUE_AMOUNT, PAYMENT_STATUS • DUE_DATE (DATE)	LIBRARY_BOOKS & Transactions • BOOK_ID (PK, NUMBER) • TITLE, AUTHOR, ISBN • TOTAL_COPIES, AVAILABLE_COPIES • BOOK_TRANSACTIONS (Issue, Return, Due Dates)